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Key Points:

- We estimate Hall and Pedersen conductance uncertainties using global auroral satellite images from Imager for Magnetopause-to-Aurora Global Exploration (IMAGE) for an isolated substorm
- We calculate uncertainties in ionospheric reconnection electric fields related to conductance, open-closed boundary (OCB) motion, and convection
- We find that the main contributor to the reconnection electric field estimates after onset is the OCB motion

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A Quantitative Analysis of the Uncertainties on Reconnection Electric Field Estimates Using Ionospheric Measurements

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Abstract Calculating the magnetic flux transfer across the open-closed boundary (OCB) per unit time and distance—the reconnection electric field—is an important means of remotely monitoring magnetospheric dynamics. Ground-based measurements of plasma convection velocities together with velocities of the OCB are commonly used to infer reconnection rates. However, this approach is limited by spatial coverage and often lacks robust uncertainty quantification. In this paper, we assimilate Super Dual Auroral Radar Network convection measurements, ground magnetometer data, and estimates of the conductance derived from the Imager for Magnetopause-to-Aurora Global Exploration satellite imagers, using the Local mapping of polar ionospheric electrodynamics (Lompe) framework over a region in North America. We present a new method to assess various contributions to uncertainties in the derived reconnection electric fields, including a novel approach to estimate uncertainties in conductance from global auroral imaging. Our method is demonstrated on a substorm event with an associated pseudobreakup during a period of favorable observational coverage. In this case study, the uncertainties in the reconnection electric field are $\sim 5\text{--}10$ mV/m at the peak of substorm expansion, roughly 15% of the peak reconnection electric field. We find that the main contributor to the reconnection electric field estimates after substorm onset is the OCB motion, whereas during the pseudobreakup the main contributor is ionospheric plasma convection.

Plain Language Summary The solar wind is a stream of particles and magnetic field flowing away from the Sun. Under certain orientations of the solar wind magnetic field, magnetic reconnection can occur between the magnetic fields of the solar wind and Earth. Magnetic reconnection is the process by which oppositely directed magnetic field lines “break,” then merge into a new configuration. Magnetic reconnection is the primary process responsible for coupling the solar wind energy to the Earth's magnetic field, and is ultimately the principal driver of space weather. In the Earth's magnetosphere, reconnection is associated with topological changes at both dayside and nightside, where stored magnetic energy is released through a global reconfiguration of the magnetosphere which is referred to as magnetospheric substorms. We can remotely monitor this reconnection from the ground by studying the transfer of magnetic flux across the boundary between open and closed magnetic field lines. In this study, we combined data from ground-based radars, ground magnetometers, and space-based auroral imaging to monitor the reconnection rate across the entire nightside auroral oval for a full substorm event. We additionally derived uncertainties in this reconnection rate stemming from uncertainties in the ionospheric velocities, the ionospheric conductances, and the open-closed boundary motion.

1. Introduction

Magnetic reconnection in Earth's magnetosphere changes the magnetic topology from one domain (e.g., open, one foot point connected to Earth) to another (e.g., closed, both footpoints connected to Earth) and is a key driver of dayside and nightside magnetospheric dynamics. Measuring how the reconnection rate varies in magnitude, location, and time provides information about these topological changes (A. T. Aikio et al., 2006; Chisham, Freeman, et al., 2008; S. Milan, 2015, and references therein). The reconnection rate is equivalent to the reconnection electric field (V/m), and defined as the rate at which magnetic flux crosses the open-closed boundary (OCB) per unit time and distance (Vasyliunas, 1984). The reconnection electric field can be determined from ionospheric observations by taking the difference normal to the OCB between the plasma convection velocity and the velocity of the OCB, which is often approximated as the poleward boundary of the auroral precipitation

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(Chisham et al., 2022). The reconnection voltage Φ (V) is defined as the line integral of the reconnection electric field E_r along the OCB (Chisham, Freeman, et al., 2008; Imber et al., 2006; Lockwood & McWilliams, 2021; S. E. Milan et al., 2003).

Several past studies have quantified the reconnection electric fields and voltages using observations of the OCB and ionospheric convection velocity (e.g., Blanchard et al., 1996, 1997, 2001; de La Beaujardiere et al., 1991; Hubert et al., 2006; Lam et al., 2006; K. Laundal et al., 2010; Østgaard et al., 2005). However, these studies were constrained by limited spatial and temporal coverage inherent to the observational data (S. Milan, 2015). For example, Hubert et al. (2006) inferred reconnection voltages using Imager for Magnetopause-to-Aurora Global Exploration (IMAGE) Far Ultra-Violet (FUV) global auroral images for OCB identification, and convection measurements from the Super Dual Auroral Radar Network (SuperDARN) for the ionospheric convection. However, only time averages of the reconnection electric field could be estimated, making its evolution difficult to assess. In addition, only a few studies have calculated uncertainties in the reconnection electric fields, and they have been limited in scope. For example, Blanchard et al. (1997) calculated the uncertainty along a single radar beam and Lam et al. (2006) calculated it over a narrow range in magnetic local time (MLT). More recently, Chisham, Freeman, et al. (2008) estimated reconnection electric fields and their uncertainties where SuperDARN global convection mapping vectors were close to the OCB determined using Polar UVI. Developing a robust approach for calculating the reconnection rate over a large spatial domain, with quantifiable uncertainties, is a long-standing goal, but has remained a challenge.

Data assimilation-based models can mitigate the spatial limitations inherent to these observations, incorporating different types of observational data into a complete reconstruction of the ionospheric electrodynamics to a greater level of detail and/or over a larger spatial region than with individual sets of observations. Ionospheric currents, ionospheric plasma flows, and neutral winds are related through the ionospheric Ohm's law, assuming that the ionosphere is in steady-state force balance between electromagnetic and frictional forces. Kamide et al. (1981) used the ionospheric Ohm's law to obtain a partial differential equation for the electric field which can be solved when the conductance and equivalent current are known. This approach is the foundation for the Assimilative Mapping of Ionospheric Electrodynamics (AMIE) technique developed by Richmond and Kamide (1988), which can take observations from various ground-based and space-based instruments to derive patterns of ionospheric conductance, potential electric field, horizontal currents, and field-aligned currents. Boudouridis et al. (2021) used the AMIE technique to estimate the ionospheric convection velocities, with OCB velocities determined from Polar Ultra-Violet Imager (UVI) observations, to study how reconnection rates respond to dynamic pressure fronts in the solar wind.

More recently, K. Laundal et al. (2022) and Hovland et al. (2022) developed the Local mapping of polar ionospheric electrodynamics (Lompe) technique, utilizing Spherical Elementary Current Systems (SECSs) (Amm, 1997). Similar to AMIE, Lompe accepts multiple data sets and generates a self-consistent model of the ionospheric electrodynamics based on ionospheric Ohm's law. The SECS method allows for regional maps with high spatial resolution given good data coverage, an advantage over global basis functions such as spherical cap harmonics (Amm & Viljanen, 1999; D. Green et al., 2007; Torta, 2020).

A key input to ionospheric empirical models such as AMIE and Lompe are ground magnetic field data. Ground magnetic field data have been used for decades to infer ionospheric electrodynamics and changes in the auroral oval morphology (e.g., Friis-Christensen & Wilhjelm, 1975; Kamide et al., 1981; Popov et al., 2001; Pulkkinen et al., 2003). However, the interpretation of ionospheric currents derived from ground magnetic perturbations strongly depends on the conductances (e.g., E. D. Cousins et al., 2015; Gjerløv & Hoffman, 2000; K. M. Laundal et al., 2015; McGranaghan et al., 2016; Ridley et al., 2004; Weygand et al., 2012, and references therein). As a consequence, the reliability of ionospheric electrodynamics reconstructed solely from magnetometer measurements is intrinsically tied to the accuracy of ionospheric conductance estimates (Amm et al., 2008; E. D. Cousins et al., 2015; Ridley et al., 2004; Waters et al., 2015). The difficulties in obtaining accurate ionospheric conductances, and the uncertainties associated with those conductances, is a well-known problem (Waters et al., 2015). Additionally, induced telluric currents also affect the accuracy of calculating ionospheric perturbation magnetic fields from ground magnetic field measurements (Juusola et al., 2020), in some cases altering the amplitude by up to 40% (Tanskanen et al., 2001).

Ionospheric conductances can be calculated via empirical models or inferred through observations. Examples of empirical models are the Hardy model (D. A. Hardy et al., 1987), OVATION Prime model (P. Newell et al., 2009;

P. T. Newell et al., 2010; P. Newell et al., 2014), and the McGranaghan et al. (2016) and Hatch et al. (2023) models. Empirical conductance models are not well-suited for capturing temporal and spatial variations of the Hall and Pedersen conductances compared to the conductance patterns derived from observations (Lummerzheim et al., 1991). Instead, observational studies often use the Robinson equations, which are empirical expressions for Pedersen and Hall conductances given as functions of precipitating electron mean energy ($\bar{E}$) and energy flux (F_E) (Carter et al., 2020; Liemohn, 2020; Robinson et al., 1987).

Ground-based auroral observations have been used to estimate $\bar{E}$ and F_E as input to the Robinson equations (Gabrielse et al., 2021; Senior et al., 2008), but are inherently limited by cloud coverage, season, and moonlight. Space-based UV images have also been used, and do not have these limitations (Coumans et al., 2004; Frey et al., 2003; Hubert et al., 2002; Lummerzheim et al., 1991; Meurant et al., 2003). Germany et al. (1997) used Polar UVI to estimate the magnitude of the incident F_E over the entire auroral zone. The three FUV cameras onboard the IMAGE satellite provided multispectral observations of auroral dynamics and allowed for improved conductance estimates over Polar UVI by adding a channel to characterize and remove the contributions from protons to Φ_E (Frey et al., 2003). The advantages of using UV images from space to infer ionospheric conductances are their large-scale coverage at high spatial and temporal resolution and their ability to work in sunlit conditions. These qualities make the UV images from space a good tool to estimate ionospheric conductance despite their lower accuracy compared to radars and particle measurements (Brekke & Moen, 1993). Although there have been some studies on sources of uncertainties in ionospheric conductance estimates obtained from satellite images (Coumans et al., 2004; Hubert et al., 2002; Meurant et al., 2003), a quantitative analysis of how these uncertainties propagate into estimates of the reconnection rates has not been given previously in the literature.

In this paper, we estimate conductances using the Robinson equations with the E_0 and F_E estimated using IMAGE FUV data with the Frey et al. (2003) technique, and derive their uncertainties from photon counting statistics. We further present a novel technique to calculate a complete map of the conductance, which we use as input to Lompe, and its uncertainty. We estimate the reconnection electric field and reconnection voltage across a large fraction of the nightside OCB while accounting for three major sources of uncertainty. First, we address how the uncertainty in the ionospheric convection velocities, quantified in terms of Lompe model variance, propagates into an uncertainty in the reconnection electric field, assuming that the conductance is known precisely. We then separately quantify how the uncertainty in the conductance maps into an uncertainty in the ionospheric plasma convection and reconnection electric field uncertainties. Finally, we examine how uncertainties in the OCB location and velocity affect the reconnection voltage.

In Section 2 we describe the data sets used in the study. In Section 3 we explain our technique for determination of the reconnection electric fields. In Section 4 we present the results, and in the last two sections we discuss these results and summarize our conclusions.

2. Data

We demonstrate our technique by applying it to an isolated substorm that occurred from 05:20 to 07:41 UT on 04 January 2001. We use satellite images of the global aurora, ground magnetometers, and ionospheric convection data. Each data set used is presented in the following subsections.

2.1. IMAGE FUV

The IMAGE satellite was launched on 25 March 2000, and provided data until December 2005. The scientific payload included three FUV cameras: the Wideband Imaging Camera (WIC), and two Spectrographic Imagers, SI12 and SI13. WIC detected wavelengths between 140 and 190 nm, which covered the N₂ Lyman-Birge-Hopfield (LBH) band emissions and N auroral emissions. SI12 detected Doppler-shifted Lyman- α emissions produced by precipitating protons at 121.8 nm, and SI13 detected atomic oxygen emissions at 135.6 nm (Frey et al., 2003; Mende et al., 2000a, 2000b). The spin period of IMAGE was ~ 123 s and auroral images were acquired during each spin. The integration time of WIC was 10 s, whereas the integration time of SI12 and SI13 was 5 s. We use images from all FUV cameras to estimate auroral conductances (see Section 3.1), and images from the WIC camera, which has the highest count rates and spatial resolution, to determine the OCB (see Section 3.5).

2.2. SuperDARN

SuperDARN is a network of more than 35 high-frequency radars operating in both hemispheres in the frequency range 10–18 MHz (Chisham et al., 2007; Greenwald et al., 1995; Nishitani et al., 2019) to study backscatter from ionospheric irregularities. Each SuperDARN radar typically scans the ionosphere through 16 beams with a 3.24° separation in azimuthal direction. Each beam covers a group range between 180 and 3,500–4,000 km with 70–110 range gates (group range cells) separated by 45 km. The backscatter echoes for each range gate are combined into autocorrelation functions (ACF), which are integrated for ≈ 3.5 s. The ACF power and phase values from each range gate are fitted with model functions by the FITACF package in the Radar Software Toolkit (SuperDARN Data Analysis Working Group et al., 2021) to determine line-of-sight Doppler velocity, signal-to-noise ratio, and spectral width. At the next stage, these fitted data are geolocated using either the standard or Chisham virtual height model (Chisham, Yeoman, & Sofko, 2008). The geolocated line-of-sight velocity data are subsequently mapped onto an equal area grid, and a spherical harmonic fit is applied to produce maps of high-latitude plasma convection (Ruohoniemi & Baker, 1998).

In this study, we used an intermediate SuperDARN data product consisting of combined gridded line-of-sight velocity data from multiple radars. The original data were processed with FITACF v2.5, geolocated using the standard virtual height model, and the first 10 range gates were removed to eliminate E-region echoes. The resulting gridded line-of-sight velocities have a time resolution of 2 min and a spatial resolution of approximately 110×110 km. These gridded line-of-sight velocity values were used as input to the Lompe inversion code, which will be introduced in more detail in Section 3.3. We used data from up to 7 SuperDARN stations (Goose Bay, Kapuskasing, Kodiak, Prince George, Pykkvibaer, Saskatoon, and Stokkseyri).

2.3. Ground Magnetometers

We use the 1-min resolution data from a selection of 50–52 ground magnetometers (depending on availability) within the analysis region as input for Lompe. We obtained the data from SuperMAG, where the baseline is removed using the Gjerloev (2012) technique.

2.4. Solar Wind and IMF Observations

We use OMNI for solar wind and Interplanetary Magnetic Field (IMF) observations. These measurements are propagated from near L1 to the nose of Earth's bow shock, and the data set also includes a variety of geomagnetic indices (King & Papitashvili, 2005). The indices and the solar wind parameters presented in this paper are taken from the 1-min OMNI data set (Papitashvili & King, 2020).

3. Methodology

The reconnection electric field, E_r , is defined as the magnetic flux crossing the OCB per unit time per unit length, and can be written as (Vasyliunas, 1984)

$$E_r = [(\mathbf{v} - \mathbf{u}) \times \mathbf{B}] \cdot \hat{\mathcal{E}}, \quad (1)$$

where $\mathbf{v}$ is the velocity of the ionospheric convection at the OCB, $\mathbf{u}$ is the velocity of the OCB, $\mathbf{B}$ is the magnetic field, and $\hat{\mathcal{E}}$ is a unit vector tangent to the OCB defined in this paper to point in the direction of increasing MLT. According to Equation 1, positive reconnection electric fields represent closure of flux (net flow from the polar cap and into the auroral oval) and negative reconnection electric fields represent opening of flux (net flow into the polar cap). For simplicity, we treat the Earth's magnetic field as radial in the high-latitude ionosphere, hence we only use the upward component, B_u , of the International Geomagnetic Reference Field (Alken et al., 2021). The uncertainty due to this assumption is about a factor of $\sin(I)$, where I is the inclination of the Earth's magnetic field, and equal to $-B_u/B_0$ (Untiedt & Baumjohann, 1993). At auroral latitudes this ratio is within a few percent of unity, indicating the assumption introduces only a small error.

To calculate E_r we obtain estimates of the ionospheric convection velocity, $\mathbf{v}$, from the Lompe technique, and estimates of the OCB velocity, $\mathbf{u}$, via identification of the OCB from IMAGE WIC images. E_r is estimated with the velocities expressed in a Sun-fixed frame of reference. In Sections 3.1 and 3.2 we describe how the conductance and its uncertainty are estimated, respectively. We describe how we estimate $\mathbf{v}$ in Section 3.3, and the

associated uncertainties of $\mathbf{v}$ in Section 3.4 with two different methods. In Sections 3.5 and 3.6 we show how $\mathbf{u}$ and its uncertainty are estimated, respectively. The uncertainty propagation for the reconnection estimates is described in Section 3.7.

3.1. Ionospheric Conductances

We obtain estimates of the Hall (Σ_H) and Pedersen (Σ_P) auroral conductances using the Robinson equations (Robinson et al., 1987)

$$\Sigma_P = \frac{40\bar{E}}{16 + \bar{E}^2} F_E^{1/2}, \quad (2)$$

$$\frac{\Sigma_H}{\Sigma_P} = 0.45\bar{E}^{-0.85}, \quad (3)$$

where $\bar{E}$ is the mean energy of precipitating electrons and F_E is the precipitating electron energy flux. We obtain estimates of $\bar{E}$ and F_E from the IMAGE satellite auroral observations following the methodology of Frey et al. (2003), which is described below. Figure 1 summarizes the steps, which we now describe.

We first grid WIC, SI13, and SI12 measurements on a coarse grid with a resolution of 250×250 km due to the coarser resolution of the SI13 and SI12 camera. We then remove dayglow and proton contributions from each of the two cameras, such that

$$I_{WIC}^* = I_{WIC} - I_{WIC}^d - I_{WIC}^p, \quad (4)$$

$$I_{SI13}^* = I_{SI13} - I_{SI13}^d - I_{SI13}^p. \quad (5)$$

where I_{WIC}^* and I_{SI13}^* are the corrected WIC and SI13 intensities, respectively, after removal of contributions from dayglow (I_{WIC}^d , I_{SI13}^d , I_{SI12}^d) and proton aurora (I_{WIC}^p , I_{SI13}^p). For dayglow removal we use the algorithm by Ohma et al. (2023). For proton removal we subtract the proton contribution from SI13 and WIC using the photon counts in SI12 and the values given in Tables III, V, and VIII in Frey et al. (2003) (Figure 1a). We assume the proton energy to be 2 keV. We average the raw images and the calculated dayglow over three images to be consistent with the data window used in Lompe (6 min). Negative counts that arise from dayglow and proton removal are set to zero for all the cameras. Finally, to calculate $\bar{E}$ we use the ratio between I_{WIC}^* and I_{SI13}^* (Figure 1b) and Tables IV and VII in Frey et al. (2003).

We set a lower limit of 0.2 keV and an upper limit of 25 keV to the electron mean energy per Tables IV and VII of Frey et al. (2003). In addition, there are areas where I_{WIC}^* and/or I_{SI13}^* counts are low, primarily in the polar cap and sub auroral regions, and along the auroral boundary. Low values in SI13 are particularly problematic, since the I_{WIC}^*/I_{SI13}^* ratio becomes unstable. For grid cells with both low I_{SI13}^* counts (less than 3) and low I_{WIC}^* counts (less than 50), we set the average energy and its uncertainty to values based on characteristics of polar cap precipitation, 0.2 and 0.08 keV (40%), respectively (P. T. Newell & Meng, 1990; P. T. Newell et al., 2009). If I_{SI13}^* is low but I_{WIC}^* high, we regard the ratio as undetermined, and set the average energy arbitrarily to 1 keV with a high uncertainty of 25 keV. In all other cases, when I_{SI13}^* is high, the ratio is regarded to be well determined and the Frey et al. (2003) method is used along with the uncertainty propagation described in Section 3.2.

Once we have $\bar{E}$ defined for all time steps on the coarse grid (Figure 1c) we use 2D linear interpolation to interpolate $\bar{E}$ onto the finer (150×150 km) Lompe grid. To calculate F_E , we grid WIC counts on the same fine grid and then remove dayglow and proton contributions (Figure 1d) using the same procedure as for the coarse grid (Equation 4). Finally, we find F_E on the fine grid (Figure 1e) by using Table IV in Frey et al. (2003) with the estimated electron energy and I_{WIC}^* intensity.

With these estimates of $\bar{E}$ and F_E we then calculate the Hall and Pedersen conductances according to Equations 2 and 3 (Figure 1f). We additionally add a background conductance (sometimes referred to as “starlight conductance”) of 0.5 mho to the Hall conductances and 0.25 mho to the Pedersen conductances (Ridley et al., 2004). Solar extreme UV produced conductance could also be added, but is negligible for this event.

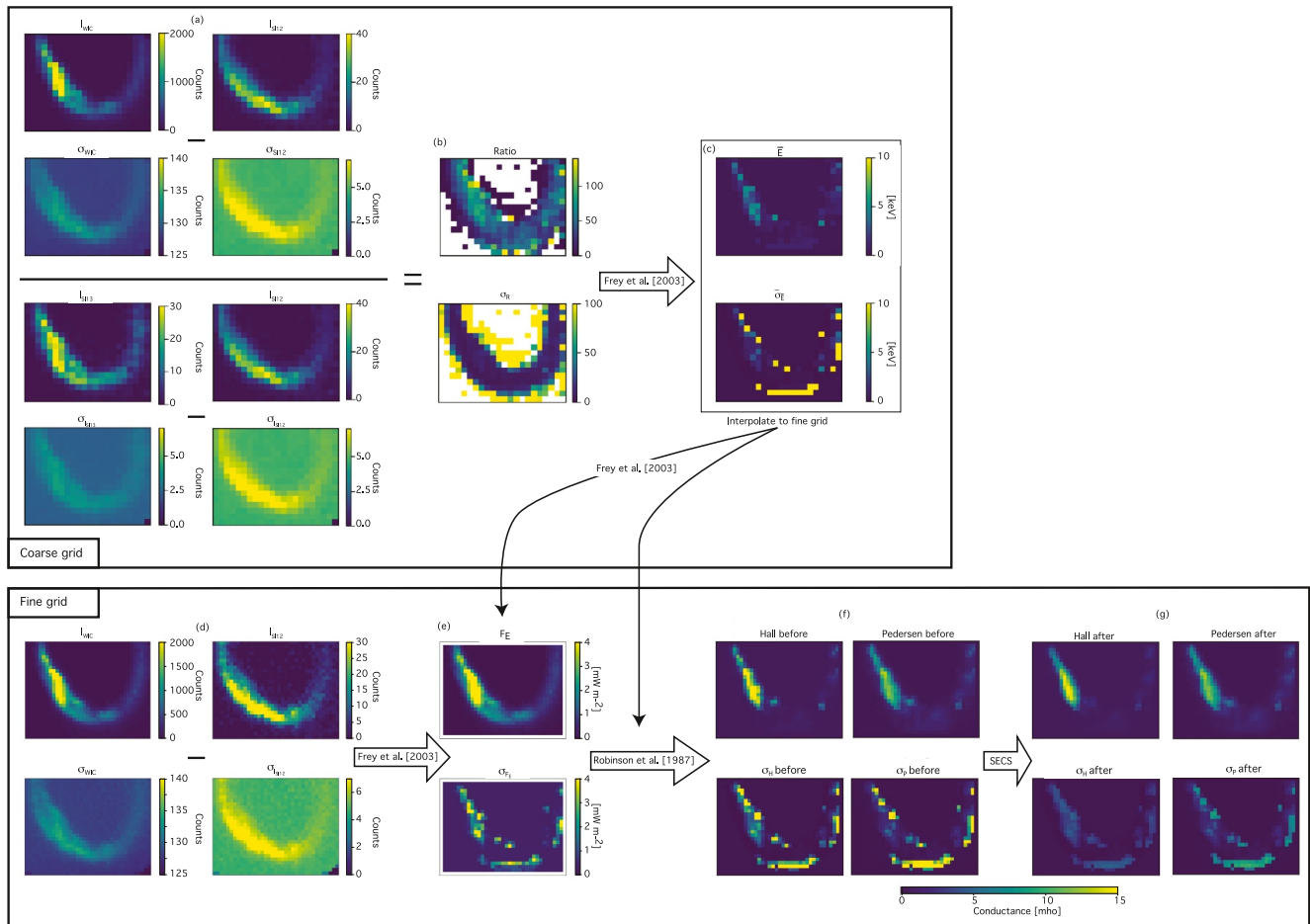


Figure 1. Summary of the steps to produce ionospheric conductances and uncertainties from the Imager for Magnetopause-to-Aurora Global Exploration data. The gridded data represent the fixed Lompe grid (same as in Figures 2–4). The grid covers the entire nightside auroral zone, with dawn toward left and dusk to right, and local midnight toward the bottom left, in each grid.

3.2. Uncertainty in the Ionospheric Conductances

We derive uncertainties in the Hall and Pedersen conductances by propagating the uncertainties in the IMAGE FUV count rates from all three cameras. The propagation of uncertainties is described in detail in Appendix A and is based on the assumption that the variance in WIC, SI13, and SI12 counts is equal to the number of counts (Poisson statistics). In some areas with low I_{WIC}^* counts, primarily the polar cap and subauroral regions, $F_E = 0$. This leads to undefined Hall and Pedersen conductance uncertainties by virtue of the factor of F_E in the denominator of Equations A10 and A11. In these areas we set the Hall and Pedersen conductance uncertainties from particle precipitation to be 40% of the conductance values to be consistent with the uncertainty in $\bar{E}$.

These conductance maps have grid cells of poorly defined conductances with high uncertainty. To address the inherently noisy auroral data, we smooth the conductances and their uncertainties using an inversion scheme based on the SECS technique (using the same formulation as the electric potential representation in Lompe), to provide robust, smoothed conductance maps to the Lompe data assimilation. Within cells with highly uncertain conductance estimates due to low count rates, this inversion uses information from neighboring grid cells to produce reasonable estimates of the conductances and their uncertainties. In other words, this inversion sacrifices spatial resolution of the conductance maps in favor of obtaining reasonable estimates of the conductances and their uncertainties over the entire model grid (Figure 1g). This general approach is flexible and applicable for data products derived from global and ground-based imagers.

For this inversion we use the same type of regularization as with Lompe (see Section 3.3.2). The values for this inversion are $\lambda_1 = 10^{-7}$ and $\lambda_2 = 10^{-6}$. We use the conductances given by this intermediate inversion as input to the Lompe inversion for the ionospheric velocities. From the conductance inversion, we use the posterior model covariance matrix projected into conductance space as the final uncertainty for the conductances.

3.3. Ionospheric Convection Velocity

Lompe assimilates SuperDARN and SuperMAG measurements to produce a map of the local electrodynamics via inversion, given the Hall and Pedersen conductances derived in Section 3.1. The Lompe analysis grid in our study is fixed in geographic coordinates to a region over North America. The IMAGE WIC spatial resolution constrains the spatial resolution of the conductance maps and the calculation of $\mathbf{u}$ (see Section 3.5) to about $\sim 1^\circ$ latitude. Therefore, we set the Lompe grid resolution to 150×150 km. Since the reconnection electric field is evaluated along the OCB, and not at regular locations relative to the Lompe grid, the Lompe output velocities at the OCB location are smoothed with a 1D Gaussian filter of similar size as the Lompe analysis grid (150 km) to avoid small-scale variations in the evaluated velocity due to the granular nature of the SECS representation (Amm, 1997). Figure 2 shows the Lompe input and output for one time step of the event, at 07:19 UT. Panel c is showing the Lompe output velocities (black) and equipotentials (blue contours) together with the SuperDARN line-of-sight velocities input to Lompe (orange).

3.3.1. Data Input to the Lompe Inversion

The inversion requires specification of the uncertainties of the convection and ground magnetic field measurements. We consider 10 nT to be appropriate estimates of the uncertainty of SuperMAG (Waters et al., 2015) and 50 m/s for SuperDARN measurements. The latter is the approximate mode of the distribution of the standard deviations of the line-of-sight velocity values in each SuperDARN grid cell. We include as a low latitude boundary condition a set of synthetic zero-flow data points at 60° magnetic latitude (MLAT) and below, similar to the use of an Heppner-Maynard boundary (Fogg et al., 2020; Heppner & Maynard, 1987; Shepherd & Ruohoniemi, 2000). We perform Lompe inversions based on data from a 6 min window centered on the times of each of the WIC images.

3.3.2. Regularization

The Lompe model output depends on two regularization parameters, λ_1 and λ_2 , that address the problem of overfitting (K. Laundal et al., 2022). The former scales the zeroth-order Tikhonov regularization which favors model solutions with smaller amplitudes, and tends to smooth the model predictions. The latter limits the amount of variation in the magnetic east-west direction of the Lompe model amplitudes. We set the values of the regularization parameters such that the Lompe output spatial resolution, determined in the same way as in Madelaire et al. (2023), matches the effective OCB resolution of ~ 0.3 hr of MLT (a few 100 km at the OCB latitudes; see Section 3.5). We then visually inspected all time steps to ensure that the model produced realistic predictions. This led to the choice of $\lambda_1 = 10^{-2}$ and $\lambda_2 = 10^{-3}$. We keep the Lompe regularization parameters constant for all inversions throughout the event.

3.4. Uncertainty in the Ionospheric Convection Estimates

3.4.1. Lompe Model Variance

The Lompe posterior model covariance matrix contains the variances (diagonal elements) and covariances (off-diagonal elements) of the model parameters. The posterior model covariance matrix represents the spread in the distribution of model parameters, and is affected by the measurement uncertainty, conductance, the grid, and especially the regularization parameters. The posterior covariance matrix does not provide an actual uncertainty but represents the variability of the model given the available information. For example, setting the regularization parameters extremely high would lead to zero model variance since the model parameters would be forced to zero regardless of the data. However, zero model variance would not mean that the uncertainty is zero, but would rather be an indication about the stability of the model.

We use the Lompe posterior model covariance matrix to calculate the variance in the convection velocity components predicted by Lompe. The posterior covariance matrix is projected into the convection data space,

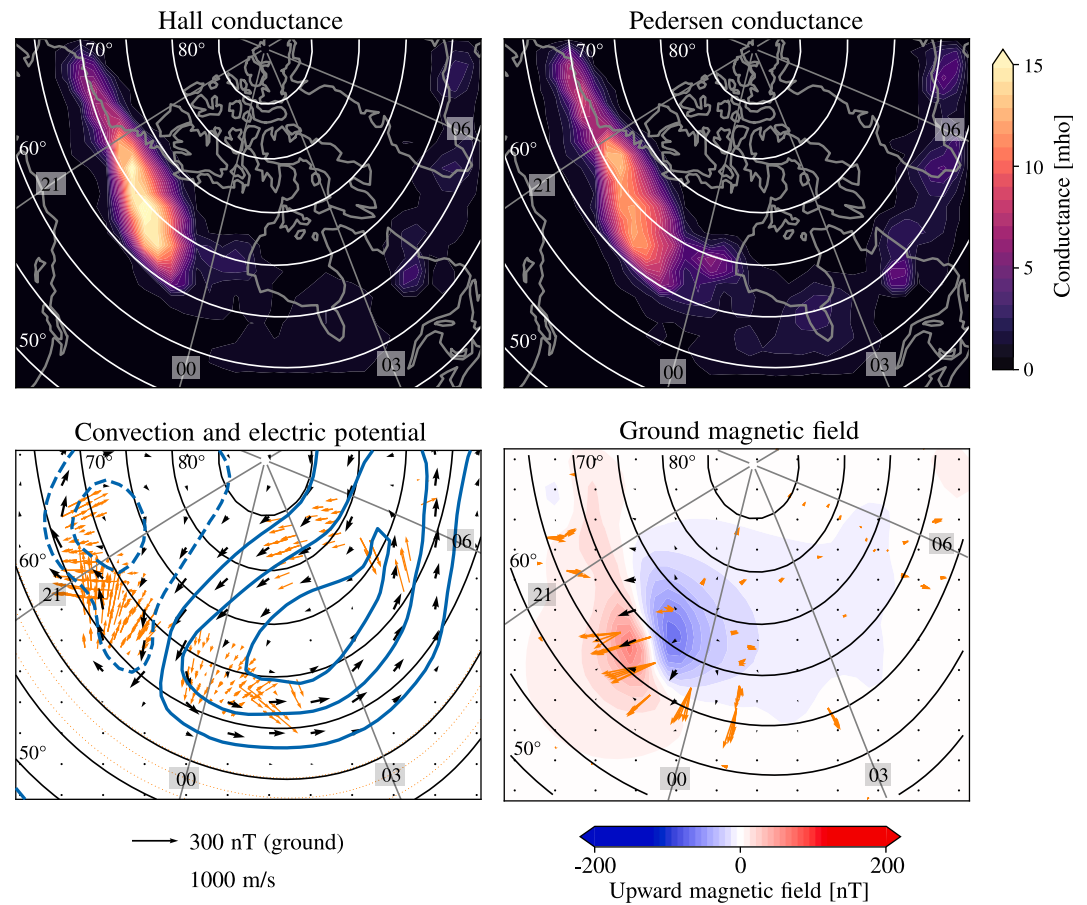


Figure 2. Lompe input and output for one time step of the event (07:19 UT) in the same grid and coordinate format as Figures 3 and 4. From left to right the first and second panels show the Hall and Pedersen conductances derived from Imager for Magnetopause-to-Aurora Global Exploration, respectively. The third panel shows the convection velocities (black arrows, Lompe output) and equipotential lines for the electric potential (blue contours, Lompe output) that result when both Super Dual Auroral Radar Network (orange arrows, input to Lompe) and SuperMAG measurements (orange arrows, input to Lompe) are used in the inversion. Positive and negative potential contours are indicated by solid and dashed lines, respectively, at intervals of 6 kV. The fourth panel shows the horizontal ground magnetic field (black arrows, Lompe output) and the upward ground magnetic field from Lompe (blue negative, red positive).

allowing us to translate these covariance values into velocities and, with the caveats above, the square root of its diagonal is hereafter interpreted as an uncertainty, and referred to as $\sigma_{v_e}^{(Lompe)}$ or $\sigma_{v_n}^{(Lompe)}$ for east and north components, respectively. Figure 3 shows $\sigma_{v_e}^{(Lompe)}$ and $\sigma_{v_n}^{(Lompe)}$ for the same time as Figure 2. These quantities are smallest (light red) where the SuperDARN data constrain the model, and largest (dark red) where there is lack of observations, meaning that the model parameters are most uncertain where the data coverage is poorest. For example, in the upper right cluster of measurements in the two panels of Figure 3, it is clear that the line-of-sight of that radar is in the eastward direction (see also Figure 2c), since the variance in the northward direction remains high. Because Lompe and similar techniques are limited in their ability to predict convection patterns in regions with no data coverage, one should be cautious when drawing conclusions about the physical behavior in areas with no data coverage. Shepherd and Ruohoniemi (2000) were the first ones to demonstrate how no/sparse data coverage influence the convection patterns in the context of SuperDARN convection mapping. This has also been discussed later on by E. Cousins et al. (2013).

3.4.2. σ , Due To Conductance Uncertainties, Given a Current System

We can use the Lompe model output to estimate the uncertainty in the ionospheric convection due to the uncertainty in the conductance. In the Lompe technique, conductance is treated as being precisely known. As such,

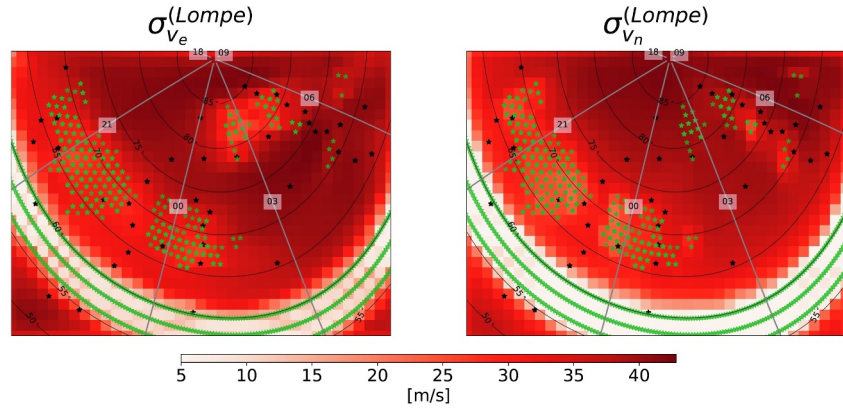


Figure 3. Lompe model variance projected into velocity space for one time step (07:19 UT). The left panel shows $\sigma_{v_e}^{(Lompe)}$ and the right panel shows $\sigma_{v_n}^{(Lompe)}$. Green markers indicate the location of Super Dual Auroral Radar Network observations and the zero-flow boundary condition, and black markers indicate the location of SuperMAG observations.

uncertainties in the conductance are not incorporated into the model variance mentioned earlier. However, we can estimate the magnitude of the uncertainty in the ionospheric convection (and consequently uncertainty in E_r) created by uncertainties in the conductance by applying the ionospheric Ohm's law, discussed next.

Conductances, velocities, and currents are linked by the ionospheric Ohm's law, which is solved here for the components of the electric field

$$E_e = \frac{J_e \Sigma_P - J_n \Sigma_H}{(\Sigma_P^2 + \Sigma_H^2)}, \quad (6)$$

$$E_n = \frac{J_n \Sigma_P + J_e \Sigma_H}{(\Sigma_P^2 + \Sigma_H^2)}, \quad (7)$$

where J_e and J_n are the ionospheric current in the east and north directions, and Σ_P and Σ_H are the Pedersen and Hall conductances. Based on uncertainties in the Hall and Pedersen conductances (σ_{Σ_P} and σ_{Σ_H}), we derive an expression for σ_{E_e} and σ_{E_n} using Equation A1, under the assumption that the ionospheric current has no uncertainty, such that

$$\sigma_{E_e} = \sqrt{\left(\frac{J_e(\Sigma_P^2 + \Sigma_H^2) - 2J_e \Sigma_P^2 + 2J_n \Sigma_H \Sigma_P}{(\Sigma_P^2 + \Sigma_H^2)^2} \right)^2 \sigma_{\Sigma_P}^2 + \left(\frac{-J_n(\Sigma_P^2 + \Sigma_H^2) + 2J_n \Sigma_H^2 - 2J_e \Sigma_H \Sigma_P}{(\Sigma_P^2 + \Sigma_H^2)^2} \right)^2 \sigma_{\Sigma_H}^2}, \quad (8)$$

$$\sigma_{E_n} = \sqrt{\left(\frac{J_n(\Sigma_P^2 + \Sigma_H^2) - 2J_n \Sigma_P^2 - 2J_e \Sigma_H \Sigma_P}{(\Sigma_P^2 + \Sigma_H^2)^2} \right)^2 \sigma_{\Sigma_P}^2 + \left(\frac{J_e(\Sigma_P^2 + \Sigma_H^2) - 2J_e \Sigma_H^2 - 2J_n \Sigma_H \Sigma_P}{(\Sigma_P^2 + \Sigma_H^2)^2} \right)^2 \sigma_{\Sigma_H}^2}. \quad (9)$$

Using $\mathbf{v} = \mathbf{E} \times \mathbf{B}/B^2$ we obtain the following uncertainties in the convection components:

$$\sigma_{v_e} = \sqrt{\left(\frac{\sigma_{E_n}}{B} \right)^2}, \quad (10)$$

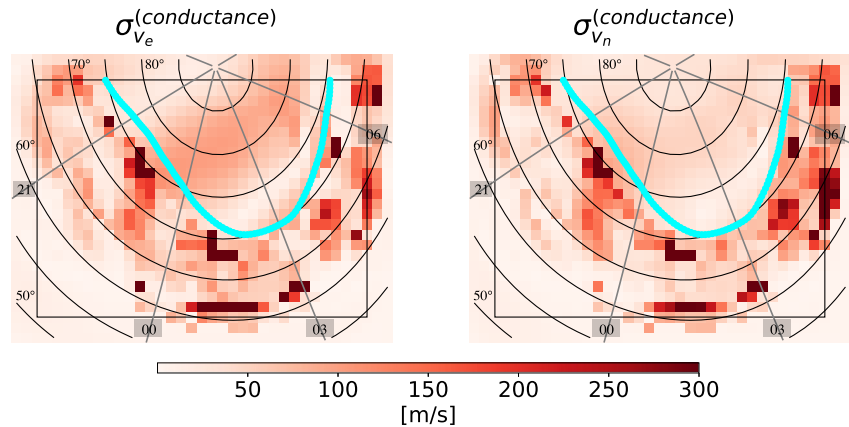


Figure 4. Uncertainty in the eastward (left panel) and northward (right panel) ionospheric convection velocities due to conductance for one time step (07:19 UT). The cyan points indicate the location of the open-closed boundary.

$$\sigma_{v_n} = \sqrt{\left(\frac{\sigma_{E_e}}{B}\right)^2}. \quad (11)$$

Thus, by utilizing the ionospheric currents from Lompe with the conductance uncertainties derived in Section 3.2, we calculate the uncertainty in ionospheric convection velocity. Figure 4 shows the uncertainty distribution for the ionospheric velocities for the same time step (07:19 UT) as in Figures 2 and 3 using Equations 10 and 11. The uncertainty in the convection velocity is overall low around the OCB and in the subauroral regions, following from the conductance uncertainties (see Section 3.1). J is assumed to be exact in these calculations and they therefore give only an order of magnitude estimate of the conductance-related uncertainty in the velocity estimates. We discuss some of the caveats with this approach, and suggest a possible alternative, in Section 5.1.2.

3.5. Velocity of the OCB

The velocity of the OCB is determined from the temporal and spatial evolution of the OCB. The OCB location is identified in the dayglow-corrected auroral images from the IMAGE satellite's WIC camera, with the assumption that the poleward edge of the auroral oval observed in WIC is the transition from closed to open magnetic field lines (Chisham et al., 2022). We manually draw the OCB by tracing an approximate contour of constant intensity, set to 50 counts. Slightly changing this threshold did not substantially affect our results, in accordance with the findings of Østgaard et al. (2005). The boundaries are then linearly interpolated and smoothed using a Savitzky-Golay filter to an equal spacing of 0.1 MLT. This produces a 2D array of boundary latitudes as function of MLT and time. To overcome the IMAGE satellite wobbling as a result of an antenna shortening (J. Green & Reinisch, 2003), we additionally smoothed the OCB using a 2D Gaussian filter with a standard deviation of 1 hr MLT and 2 min in time. With this smoothing applied, the effective spatial resolution of the OCB determined empirically becomes ~ 0.3 hr of MLT.

Given the OCB latitude (θ_b) as a function of MLT (ϕ) and time, the following analytic expressions provide the OCB velocity components normal to the OCB, u_θ and u_ϕ (Hatch, 2023; Ohma et al., 2024):

$$u_\theta = r \frac{\partial \theta_b}{\partial t} \frac{\sin^2 \theta_b}{\sin^2 \theta_b + \left(\frac{\partial \theta_b}{\partial \phi}\right)^2}, \quad (12)$$

$$u_\phi = -\frac{r}{\sin \theta_b} \frac{\frac{\partial \theta_b}{\partial t} \frac{\partial \theta_b}{\partial \phi}}{1 + \left(\frac{\partial \theta_b}{\partial \phi}\right)^2}. \quad (13)$$

3.6. Uncertainty of the OCB Velocities

The uncertainty of the OCB velocities is estimated using Equations B1 and B2 in Appendix B via the `auroralbndrytools` Python package (Hatch, 2023). The uncertainty of the OCB velocities stems entirely from uncertainties in the identified locations of the OCB identified in the WIC images. We assume that the OCB location cannot be determined to a better resolution than the pixel resolution at apogee, 120 km or $\sim 1^\circ$ latitude, according to Table III in Mende et al. (2000b). We therefore use $\sigma_{\theta_b} = 1^\circ$ as the uncertainty in θ_b .

Two other important quantities that are needed in the propagation of the OCB velocity uncertainty are the uncertainty in the variation of θ_b with respect to time, and the uncertainty in the variation of θ_b with respect to space (ϕ /MLT direction). Approximating the derivatives as first-order finite differences, these are given by

$$\frac{\partial \theta_b}{\partial t} = \frac{\theta_b(t+1) - \theta_b(t)}{\Delta t} = \frac{\theta_b(t+1)}{\Delta t} - \frac{\theta_b(t)}{\Delta t}, \quad (14)$$

$$\sigma_{\frac{\partial \theta_b}{\partial t}} = \sqrt{\left(\frac{1}{\Delta t}\right)^2 \sigma_{\theta_b}^2 + \left(\frac{-1}{\Delta t}\right)^2 \sigma_{\theta_b}^2 - 2 \frac{\rho_t \sigma_{\theta_b}^2}{\Delta t^2}} = \sqrt{2 \frac{\sigma_{\theta_b}^2}{\Delta t^2} (1 - \rho_t)}. \quad (15)$$

For Equation 15 the correlation coefficient ρ_t represents the correlation between the OCB locations at different time steps. To estimate ρ_t we take the OCB latitude location for each MLT as a function of time, and correlate that with the latitudes from the same MLT but offset by 1 time step. The median correlation coefficient ρ_t is 0.994.

The same approach is used to calculate $\partial \theta_b / \partial \phi$ and its uncertainty. The corresponding correlation coefficient ρ_ϕ , represents the correlation between the OCB latitude at two adjacent longitudes at the same time step. To estimate ρ_ϕ we take the OCB latitude at each location, for all times, and correlate these with the OCB latitude at $\Delta \text{MLT} = 0.3$ (the effective spatial resolution of the OCB), and then take the median of all these correlations. The median correlation coefficient ρ_ϕ in this case is 0.995. We use $\sigma_{\frac{\partial \theta_b}{\partial t}}$ and $\sigma_{\frac{\partial \theta_b}{\partial \phi}}$ as input to Equations B1 and B2.

3.7. Uncertainty of the Reconnection Estimates

Uncertainty in the reconnection electric field is calculated through propagation of uncertainties in Equation 1. Neglecting the uncertainty related to B_u , t_e , and t_n , the uncertainty in the reconnection electric field is

$$\sigma_{E_r} = B_u \sqrt{t_e^2 (\sigma_{v_e}^2 + \sigma_{u_n}^2) + t_n^2 (\sigma_{v_e}^2 + \sigma_{u_e}^2)}, \quad (16)$$

where σ_{v_e} and σ_{v_n} are the uncertainties in the convection velocities, and σ_{u_e} and σ_{u_n} are the uncertainties in the OCB velocities. We smooth σ_{v_e} and σ_{v_n} with a 1D Gaussian filter to be consistent with the smoothing done on v_e and v_n . In Sections 4 and 5 we examine the uncertainties from both $\mathbf{u}$ and $\mathbf{v}$ separately to investigate their relative importance in estimates of reconnection rates and voltages. The uncertainty of the reconnection voltage across the entire nightside is then the sum of the reconnection rate uncertainties from each segment along the OCB. This sum is an upper limit for the reconnection voltage uncertainty, given the uncertainties of the reconnection rate of each segment.

4. Results

4.1. Event Overview

We analyze the time period 05:20–07:41 UT on 04 January 2001, during which a pseudobreakup and a magnetospheric substorm was observed by IMAGE, SuperDARN, and ground magnetometers. Figure 5 shows the IMF and geomagnetic indices 3 hr before, during, and 3 hr after this time interval. Figure 5c shows the dayside reconnection voltage, Φ_D , calculated using the S. Milan et al. (2012) coupling function, which is based on upstream solar wind parameters. Figure 5d (SYM-H index) and Figure 5e (AL index) indicate this was a relatively isolated substorm during a geomagnetically quiet period, with a minimum SYM-H value around -22 nT. Figure 5f shows the evolution of the OCB in the analysis region between approximately 18–5 MLT, where the color represents the MLAT of the OCB. Thick red lines mark the beginning and the end of our analysis window and the thin red line in Figures 5e and 5f indicates the substorm onset time according to the Frey et al. (2004)

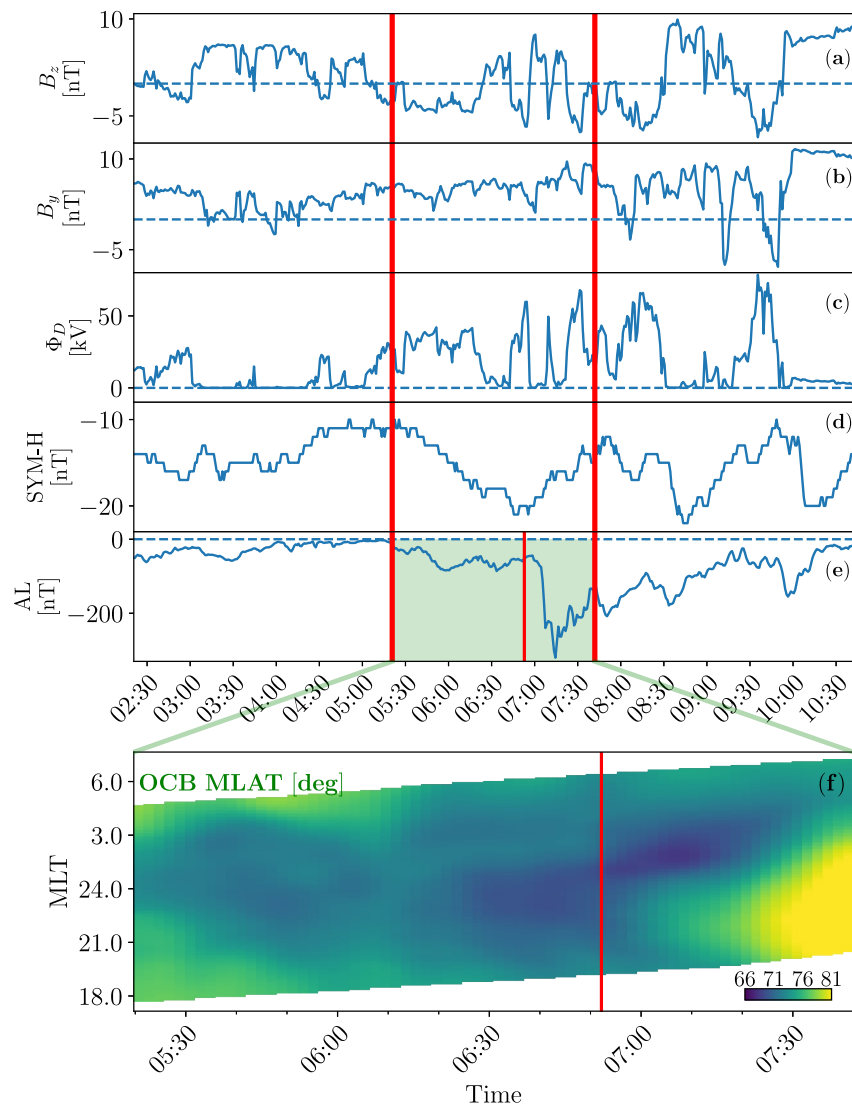


Figure 5. Solar wind parameters and geomagnetic indices 3 hr before, during, and 3 hr after for the case study event on 04 January 2001, and the progression of the open-closed boundary (OCB) over the event. Shown from top to bottom are IMF B_z and B_y , Φ_D (the S. Milan et al., 2012, coupling function), the SYM-H index, the AL index, and the temporal evolution of the OCB, where the color represents the magnetic latitude. Thick red lines indicate the beginning and the end of the case study event. The thin red line in panels (e) and (f) indicate substorm onset.

IMAGE substorm list. The range of visible MLTs in Figure 5f changes throughout the event, reflecting that the Lompe grid is fixed in geographic coordinates (this effect is also visible in Figures 7 and 8). The solar wind speed (not shown) ranges between 360 and 420 km/s.

The onset of the substorm is identified at 06:52 UT in the Frey et al. (2004) IMAGE substorm list, whereas the AL index indicates a slightly later time, approximately 07:00 UT. The IMF B_z turned southward around 05:00 UT, remained southward until around 06:22 UT, and then reversed six times throughout the remainder of the event. IMF B_y was positive throughout the event. Figure 5f shows that prior to substorm onset the OCB started to move equatorward (dark blue starting near 06:03 UT). After substorm onset the duskward OCB moved poleward, whereas the dawnward OCB moved equatorward until approximately 07:23 UT, after which it also moved poleward. This is also visible in Figure 6.

Figure 6 shows a sequence of snapshots of Lompe inputs and outputs, the motion of the OCB, and the ionospheric convection velocity at the OCB for 16 time steps. The left panels show the locations of the SuperDARN

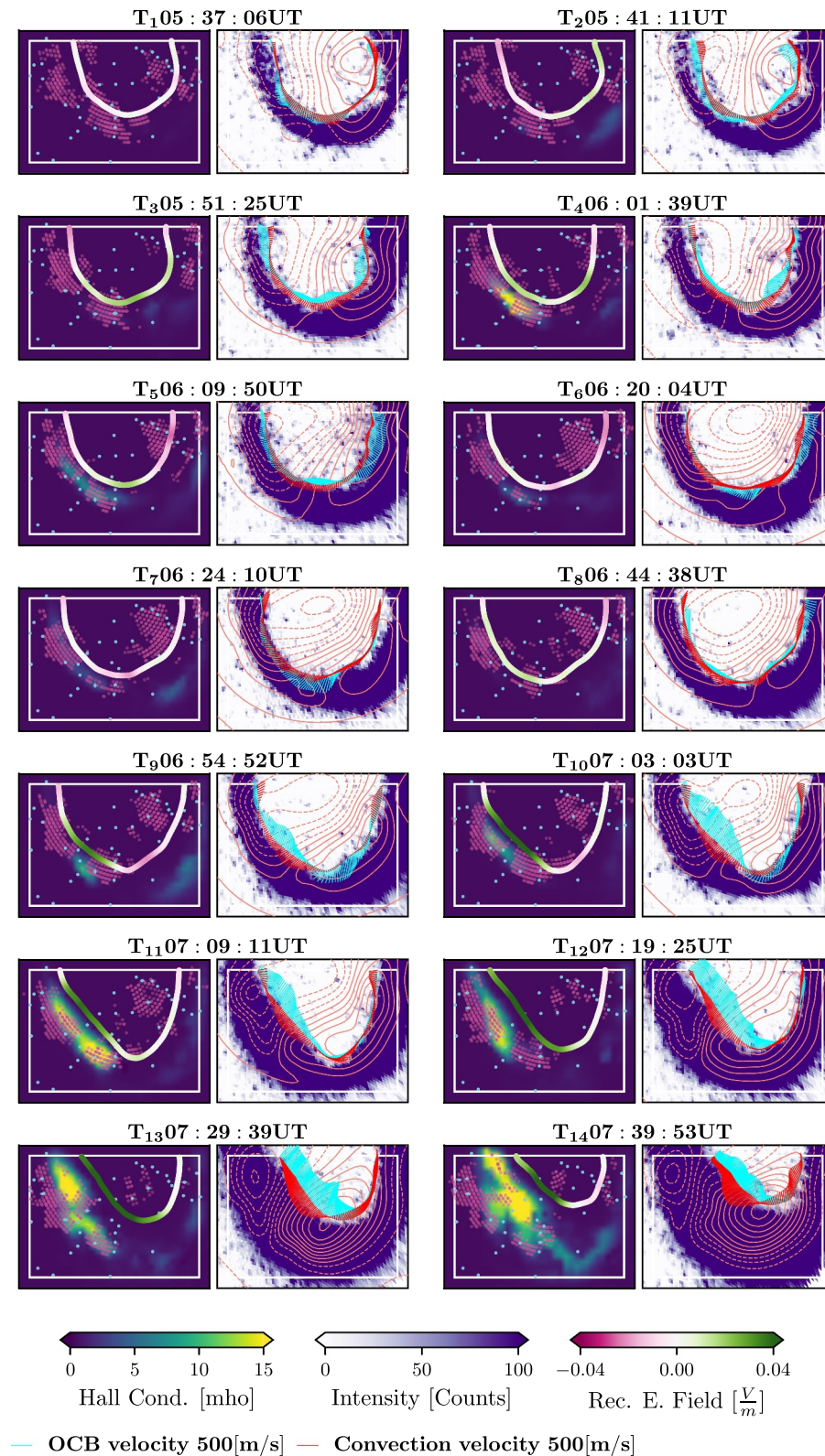


Figure 6. Evolution of the open-closed boundary (OCB) and related ionospheric electrodynamics for 16 time steps. Each subfigure consists of two panels. The left panels show the Hall conductance, the locations of the data (SuperMAG in cyan and SuperDARN in magenta), and the reconnection electric fields plotted along the OCB (pink and green indicate negative and positive respectively). The right panels show the photon counts, the equipotential lines of the electric potential at intervals of 3 kV, the OCB velocities (cyan), and the convection velocities (red).

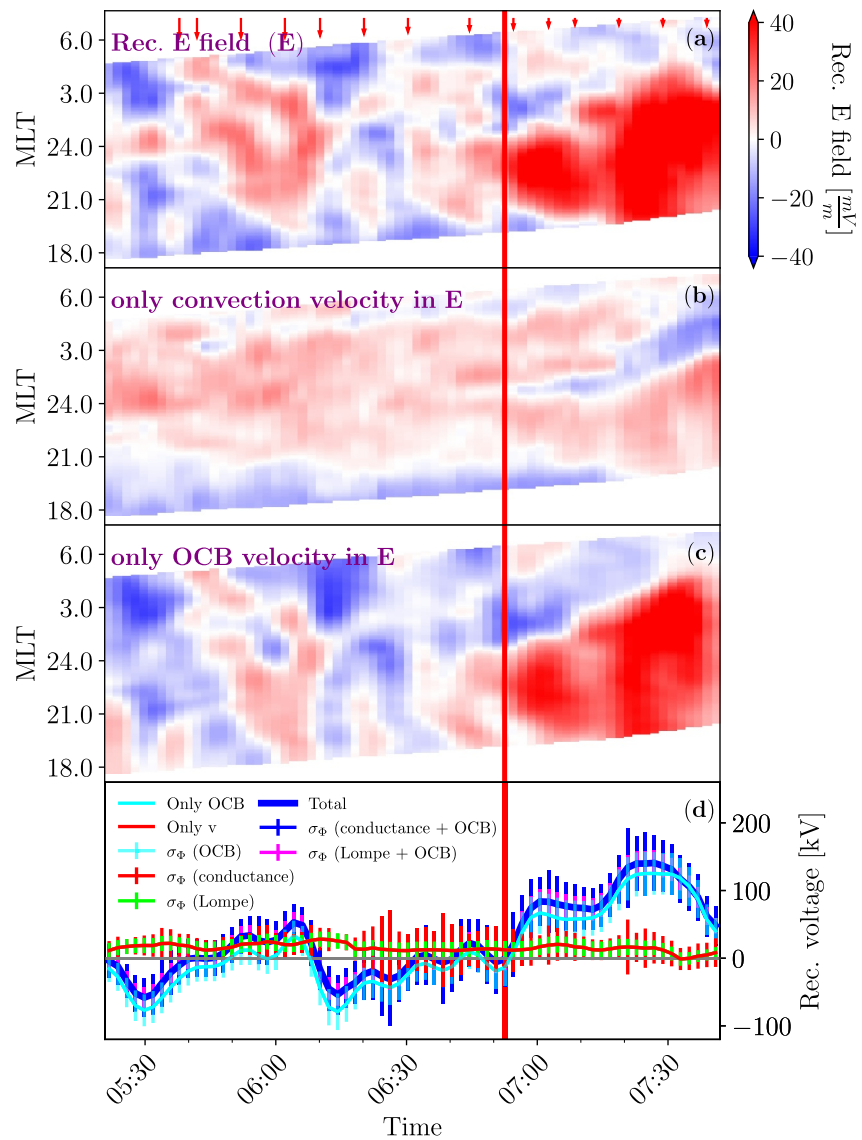


Figure 7. The contribution of the open-closed boundary (OCB) motion and the ionospheric convection to the temporal and spatial evolution of the reconnection electric field estimates. (a) Reconnection electric field including both ionospheric convection and OCB velocities in Equation 1. (b) Reconnection electric field based solely on ionospheric convection in Equation 1. (c) Reconnection electric field based solely on OCB motion in Equation 1. Time is on the x axis and magnetic local time on the y axis at the left side of the figure. (d) The reconnection voltages (kV) for the total and the two different contributions to Equation 1 with the associated uncertainties as indicated in the legend.

backscatter (magenta dots) and the SuperMAG data (cyan dots), as well as the estimated reconnection electric fields along the OCB (pink and green indicate negative and positive respectively). The reconnection electric field is only calculated within the white box, a subset of the Lompe solution region within which we consider edge effects to be unimportant. Only the Hall conductance is shown in the left panels, but both Pedersen and Hall conductances are used as input to Lompe. The right panel shows the associated aurora photon counts from the WIC camera, the velocity of the OCB (cyan quivers), and the Lompe convection velocities (red quivers). The right panels also show equipotential lines of the electric potential. Equipotential lines, convection velocities, and OCB velocities are shown as they would be seen from Earth's corotating frame of reference (i.e., by an observer standing at a fixed point on Earth).

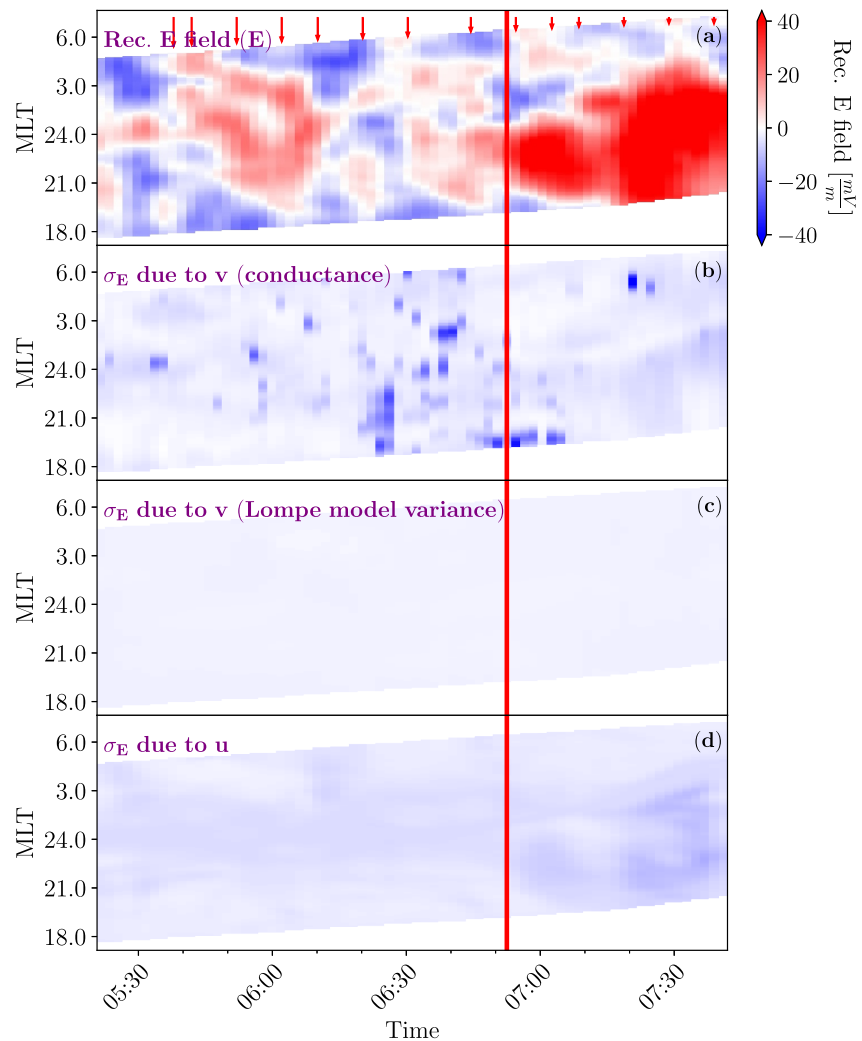


Figure 8. The contribution of different sources of uncertainty to the reconnection electric field estimates. (a) Reconnection electric field including both ionospheric convection and open-closed boundary (OCB) velocities in Equation 1. (b) Reconnection electric field uncertainty due to ionospheric convection based on uncertainty in conductance and a given current system. (c) Reconnection electric field uncertainty due to ionospheric convection uncertainty from the Lompe model variance. (d) Reconnection electric field uncertainty due to OCB velocity uncertainty. Time is on the x axis and magnetic local time on the y axis at the left side of the figure. The colorbar is the same for all panels in this figure. The uncertainties in (b)–(d) (all positive) have been multiplied by -1 to aid the comparison with the negative reconnection rate in (a). The small red arrows in the top panel indicate the time steps of the snapshots shown in Figure 6.

4.2. Reconnection Electric Field and Reconnection Voltage Estimates

Figure 7a shows the reconnection electric field (Equation 1) as a function of MLT and UT. Figure 7b shows the contribution from only the ionospheric convection assuming a stationary OCB (i.e., $\mathbf{u} = 0$ in Equation 1), and Figure 7c shows the contribution from only OCB movement assuming no plasma convection (i.e., $\mathbf{v} = 0$). Figure 7d shows in blue the integrated reconnection electric fields over the MLTs in the analysis region (reconnection voltage) for the contributions from both $\mathbf{v}$ and $\mathbf{u}$, and in red and cyan the reconnection voltage contributions from $\mathbf{v}$ and $\mathbf{u}$, respectively. It also shows their uncertainties, which are discussed in detail in Section 4.3.

Overall, the structure of the total reconnection electric field (Figure 7a) follows the structure of the reconnection electric field based only on the OCB motion (Figure 7c). In Figure 7a there is a region of slightly increased reconnection electric fields during 05:50–06:10 UT with minor poleward motion of the OCB (Figure 5f). This interval coincides with the pseudobreakup and a small decrease in the AL index. As described by Rostoker (1998),

pseudobreakups are not associated with a large scale poleward motion of the OCB and this is consistent with what is seen in Figure 6. Following substorm onset at 06:52 UT, the reconnection electric fields between approximately 18–01 MLT sharply increased, having a temporal and spatial structure that follows the poleward and dawnward motion of the OCB after substorm onset (see Figure 5f). The reconnection electric fields in Figure 7c indicate that the nightside reconnection region expanded in MLT following substorm onset, and the substorm had two distinct intensifications. The first was associated with the main substorm onset near 06:52 UT, and followed by an intensification of the substorm with an additional OCB motion near 07:20 UT.

Figure 7d shows Φ_N (blue line in Figure 7d), which is the integral of E_r in MLT at each time step and represents the total amount of magnetic flux crossing the nightside OCB per unit time. As with E_r , Φ_N also shows two distinct substorm phases following substorm onset. This temporal evolution of Φ_N is driven by the contribution of Φ_N that is based solely on OCB motion. Note that the estimates of Φ_N associated with ionospheric convection (solid red lines in Figure 7d) increased very little after substorm onset, even though there is abundant reconnection. We discuss this further in Section 5.3.

4.3. Uncertainty Analysis

Figure 8 shows E_r followed by the three sources of uncertainty we have estimated. The uncertainty related to conductance (Figure 8b) is relatively moderate, with some high-value spikes at OCB locations where the conductance is undetermined due to low SI13 counts. The contribution to the uncertainty from the Lompe model variance (Figure 8c) is overall lower compared to the other uncertainties, because the model is constrained by regularization. Finally, the uncertainty due to the OCB velocity (Figure 8d) generally higher and increases after substorm onset (06:52 UT). It is largest near the peak of the expansion phase (~07:23 UT), which is when the OCB is most dynamic over the IMAGE time scales.

The contribution to the reconnection rate uncertainty from conductance is larger than the contribution from the Lompe model variance (Figure 7d), reflecting the spikes in Figure 8b. As a consequence, the uncertainties on Φ_N are higher when including the uncertainties in velocities due to conductance.

5. Discussion

We have explored the effects of uncertainties in ionospheric parameters on calculations of the reconnection electric field. Our approach has been to isolate different sources of uncertainties and determine quantitatively how they propagate into E_r . While it would be insightful to consider the combined effect of the conductance and velocity uncertainties, such an exploration is more complex and beyond the scope of this study. It is moreover beneficial to understand the impact of conductance on its own for a clearer intuition about how the two sources of uncertainty contribute differently.

A key result of this study is that, despite our robust approach for quantifying the reconnection electric field, we observe nonphysical negative reconnection estimates that exceed our uncertainty estimates. This suggests that challenges still remain in quantifying all sources of uncertainty and error in the reconnection estimates. We note that the negative reconnection electric fields (blue) in Figure 7a follow the structure of the temporal evolution of the OCB. A few previous studies, such as Lyons et al. (1989) and de La Beaujardiere et al. (1991), have discussed the possibility of negative reconnection electric fields. Lyons and Pridmore-Brown (1992) furthered this theory using observations from the Sondrestrom radar. However, we deem opening of flux on the nightside as physically unlikely, and attribute the negative electric fields to uncertainties in either the OCB velocity or the Lompe convection velocity. These negative reconnection electric fields are generally associated with regions where the Lompe equipotentials are aligned with the OCB (hence perpendicular to the OCB velocity). For example, the large negative electric field in Figure 7a near 06:20 UT is associated with equatorward motion of the OCB and anti-sunward convection near the dawn flank as shown in Figure 6. Another example is near the end of the substorm. The images from 07:29 and 07:39 UT in Figure 6 show that the return cell convection is slightly misaligned with the OCB, and the projection of this sunward return flow onto the OCB velocity yields apparent opening of flux. The component of the ionospheric velocity along the OCB motion is what contributes to the reconnection electric fields, according to Equation 1. Therefore, even slight changes of the relative angles between the convection flow and the OCB motion and/or different vector magnitudes could lead to an unphysical opening of flux.

The observed unphysical negative reconnection electric fields generally exceed the uncertainty estimates, indicating that we have not accounted for all sources of uncertainties, despite our robust estimates. There are in fact many other sources of uncertainty that have not been accounted for in the uncertainty estimates of the reconnection electric fields, and which we discuss below.

5.1. Challenges With Quantifying the Sources of Uncertainty in the Reconnection Estimates

5.1.1. Challenges in Estimating Ionospheric Conductances and Their Uncertainties

The Frey et al. (2003) method makes multiple assumptions about how the atmosphere responds to electron and proton precipitation. For example, Frey et al. (2003) assumed that the incident electrons and protons are Maxwellian and kappa distributions, respectively, with $\kappa = 3.5$ (for $\kappa > 2.45$ there is little difference between Maxwellian and kappa distributions, see introduction in Hatch et al. (2018)), and also used a model for the altitude distribution of the atmospheric constituents O_2 , N_2 , and O . The atmospheric composition and densities at the height of the auroral emissions can change with for example, season and auroral activity. This causes deviations from the dependencies between the camera counts and the derived precipitation characteristics as provided by Frey et al. (2003), who stated that their estimates of E_0 and F_E are accurate to within $\pm 50\%$ of the true value. The contribution from uncertainties in the atmosphere modeling is however outside the scope of this paper. When we propagate our calculated uncertainties of the ionospheric conductances to uncertainty of the reconnection electric fields, using Equation 16, we obtain an approximately 15% uncertainty on E_r at the peak of substorm expansion.

In addition to these sources of uncertainty, one must also consider the validity of the Robinson equations. These equations are widely used because they enable estimates of the Pedersen and Hall conductances based on simple moments of the precipitating electron distribution function. However, they are limited to a specific energy range for the precipitating electrons, and they yield large errors in the conductances when the energy spectra of the electron precipitation departs from a Maxwellian distribution (D. Hardy et al., 2008; Liemohn, 2020). Kaeppeler et al. (2023) recently addressed this issue and derived empirical relationships for the Hall and Pedersen conductances using incoherent scatter radar and all-sky imagers. They found the median Pedersen conductance to be in good agreement with the Robinson equations, whereas they found the median Hall conductances to be larger. We have limited ourselves in this study to the uncertainties originating from the variability of the optical emissions. How much the use of the Robinson equations add to this uncertainty is an open question.

5.1.2. Applicability of the Conductance Method Uncertainty Given a Current System

We demonstrated that the uncertainty of the reconnection electric fields derived through conductances is overall larger than the Lompe variance method. Additionally, for conductances that were undetermined due to low grid cell counts, we assigned a high uncertainty, and this method led to spikes in the reconnection electric field uncertainty when the regions of high uncertainty coincide with the OCB (see Figure 8). Since the conductance method does not take into account whether there are any convection measurements close to the OCB, it neglects the direct constraints on the convection field those measurements would provide. Generally speaking, the conductance method for E_r uncertainty is more appropriate in regions with ground magnetometers and no SuperDARN convection velocities, whereas in regions with SuperDARN it is more appropriate to use the Lompe model variance. An additional point relates to the fact that the Lompe method is local. Local magnetic field perturbations could be explained by distant currents, yet our approach of relating the conductance to the velocity via the current at a single grid cell does not take into account the uncertainties in conductances at other locations that may be affecting the derived current system at that grid cell that produces a local magnetic field perturbation. This could be addressed with more advanced methods that take into account non local influences in the local magnetic field measurements, including modifying the Lompe inversion to take into account conductance uncertainties. For example, a different approach could be to repeatedly sample from a distribution of conductances with the associated uncertainties, and investigate the resulting model spread due to this distribution. Another remark is with regards to telluric induced currents, which are not taken into account in Lompe. As shown by Juusola et al. (2020), telluric induced currents also affect the ground-based magnetic field measurements, which we assume in Lompe are all due to ionospheric currents. This introduces some modest error into the ionospheric velocities, which we do not take into account in this paper.

5.1.3. Challenges With Associating the Lompe Model Variance With a σ_v

The Lompe model variance, which we related to an uncertainty in v , does not include all sources of error and includes assumptions about the physics that may introduce further error. For example, we assume the neutral wind to be zero in the ionospheric Ohm's law. A non-zero neutral wind would yield an error in the estimated ionospheric convection where the ionospheric convection predicted by Lompe is derived primarily from magnetometer data. Given the sustained ionospheric velocity observed during this interval, a non-zero neutral wind field is possible, although we have no data with which to investigate the possible contribution to the uncertainty. In Lompe, we also assume that all variations in the magnetic fields within the Lompe grid are explained by currents within the grid. If the observed magnetic field perturbations are due to sources outside the grid, this will lead to over-estimation of the electric fields. Additionally, since Lompe is a local model, the representation of a uniform background electric field will lead to slightly unrealistic features at the edge of the Lompe grid. Both of these caveats are discussed in more detail in K. Laundal et al. (2022). This is the reason why we only evaluate the Lompe model on a smaller region inside the main grid, as illustrated in Figure 6. Furthermore, the 6 min averaging window used with Lompe puts additional restrictions on the ability to fit rapid changes in the fields. In addition, induction electric fields can add up to tens of percent to the ionospheric convection electric fields Madelaire et al. (2024). However, this is most valid on short time scales ($\lesssim 5$ min), which is less than our averaging window.

One final remark goes to the uncertainties in the SuperDARN measurements. Gillies et al. (2009) and Gillies et al. (2010, 2011) discussed how the ionospheric refractive index effects the ionospheric convection velocities, leading to an underestimation of the Doppler velocities. Moreover, our data were processed with the standard virtual height model, which is known to place measurements at increasingly erroneous locations with increasing range, and more accurate virtual height models could potentially be used in the gridding of the data to reduce this uncertainty (Chisham, Yeoman, & Sofko, 2008; Thomas & Shepherd, 2022). Furthermore, even if basic attempts have been made to remove non-F-region scatter, some E-region echoes can still be present in the pre-processed data (e.g., Chisham & Pinnock, 2002). Additionally, with respect to the gridded data, a median-based filter was applied across three consecutive range cells, three consecutive beams, and three consecutive time samples. Such smoothing of the raw line-of-sight velocities can introduce additional uncertainties in the SuperDARN convection velocities (Ruohoniemi & Baker, 1998). Also, the conventional empirical procedure criteria for filtering out ground scatter echoes, (e.g., Ponomarenko et al., 2007) can still interpret some of the latter as ionospheric backscatter echoes. Finally, ionospheric and ground scatter echoes could get mixed together, leading to distortions in the estimates of the ionospheric convection velocities (e.g., Ponomarenko et al., 2008). While the quantification of contributions from these factors lies beyond the scope of the current publication, it would be desirable to perform it in the future.

Compared to the uncertainty due to conductance given a current system (shown in Figure 8b), the uncertainties derived from the Lompe model variance (Figure 8c) are more stable. This relative stability reflects that the inverse problem solved in Lompe is fairly similar over time, due to the identical grid and regularization, fairly similar data coverage and properly chosen regularization parameters.

5.1.4. Challenges in Estimating σ_u

There are a number of sources of uncertainty in the OCB determination which are not quantifiable. The first is our assumption that the nightside OCB is represented by the auroral precipitation boundary. It is possible that the topological OCB and the particle precipitation boundaries are not colocated, in which case the real topological OCB boundary would be more poleward than the particle boundary, which is always located inside the magnetically closed plasmasheet. Boakes et al. (2008), Longden et al. (2010), and Chisham et al. (2022) compared the IMAGE FUV OCB proxies with those determined from Defense Meteorological Satellite Program (DMSP) measurements and showed systematic differences in the locations. They found this offset to be a few degrees and variable with MLT and time. For time dependent offsets our OCB locations could potentially be different from the true values, hence affecting the uncertainty in the OCB velocities (see Equations B1 and B2). On the other hand, if the offsets are constant in time this would not play a role in the determination of the OCB velocity uncertainty. Another issue is the ability to identify the OCB location given by particle precipitation in the auroral imagery. The determination of the OCB was based on a manual identification of the 50-count contour in the WIC images after dayglow removal. These effects are probably largest prior to substorm onset, when precipitation is low. Finally, the spatial resolution of the WIC camera limits the precision of the OCB location to no

more than $\sim 1^\circ$. During dynamic substorm activity, the OCB can move quite rapidly and irregularly as shown in Figure 6. Combined with the cadence and spatial field of view of the IMAGE FUV cameras, and the spacecraft wobble, this motion can lead to large uncertainty in the geolocation of the OCB at each time step and, therefore, in the OCB velocity. We mitigate these effects by smoothing the OCB location, however some unaccounted for uncertainty may remain contributing to the misalignment between u and v .

5.2. Comparison of Reconnection Estimates With Previous Studies

We find that Φ_N starts to increase immediately after substorm onset and peaks at around 85 ± 46 kV (85 ± 18 kV) for the conductance method (Lompe model variance) about 10 min after onset, followed by a second peak of around 140 ± 41 kV (140 ± 18 kV) about 20 min later. The mean of the uncertainty in the reconnection voltages are 16 kV for the Lompe derived uncertainty and 36 kV for the conductance method, with corresponding standard deviations of 1 and 9 kV, respectively. The large differences in the standard deviation arise because the conductances sometimes contain large spikes in the uncertainties (see Figure 8b).

Peak reconnection voltages during a substorm can vary strongly from event to event, but our values are similar to those obtained in other studies. S. Milan et al. (2007) found average rates of 85 kV using 73 hr of observations and 25 nightside reconnection events, but noted a large variation. Hubert et al. (2006) found a reconnection electric field of ~ 150 kV, however they observed this increase ~ 12 – 14 min prior to onset due to the smoothing of the OCB determined from IMAGE SI12 images. Lam et al. (2006) estimated a peak reconnection voltage of 100 kV with 10 kV uncertainty close to substorm onset, compared to the 16 and 36 kV mean uncertainties obtained here with the two different methods. However, the Lam et al. (2006) uncertainty was obtained over a relatively narrower MLT range (3 hr between 21 and 24 MLT) compared to our study. Blanchard et al. (1997) observed an increase in the median reconnection rate 20 min after substorm onset.

During the substorm expansion phase, we find peak values of E_r of ~ 56 mV/m, with an uncertainty of ~ 7 and ~ 2 mV/m for the conductance method and the Lompe model variance, respectively (Figure 8). Using SuperDARN data for both the ionospheric velocity and OCB velocity, Lam et al. (2006) found an uncertainty of ~ 7 mV/m on the reconnection electric fields, driven primarily by the uncertainty in OCB determination. In our case both ionospheric convection and OCB motion contribute to the uncertainty, in particular after substorm onset (Figure 8). Blanchard et al. (1996) reported a peak reconnection electric field of ~ 70 mV/m and an uncertainty of 11.6 mV/m on average on the reconnection electric field inferred from Sondrestrom radar measurements. de La Beaujardiere et al. (1991), using the same radar, reported reconnection electric field magnitudes of ~ 30 – 40 mV/m during substorm expansion phase. However, both studies used limited 1D Sondrestrom incoherent scatter radar measurements, and were unable to capture local time variations. Boudouridis et al. (2021) examined the reconnection electric field during a solar wind dynamic pressure front, and found nightside reconnection electric field values up to 50 mV/m. Overall, the values of the reconnection electric field, the reconnection voltage and the uncertainties found in this study are in reasonable agreement to previously reported studies.

Figure 7d shows that the overall reconnection voltage, the blue curve including contributions from both ionospheric convection velocities and OCB velocities, is small during the substorm growth phase (6:10–6:50UT), and larger during the pseudobreakup and substorm. If we take into account the uncertainties in the reconnection voltages, this result is in general agreement with that of S. Milan et al. (2007), who state that they find no evidence for a reconnection voltage background and attribute all closure of flux to substorms and other large reconnection bursts. However, this result was for the integrated reconnection electric field. Figure 7a shows that in the pre-midnight region during the substorm growth phase there is a moderate level of reconnection electric field (~ 10 mV/m). This local time structure in the reconnection electric field does not register in the reconnection voltage, as the sum over MLTs removes all spatial differences in the reconnection electric fields. Therefore, for this event, we do observe reconnection in between events, in contrast to S. Milan et al. (2007), although it is localized to a region spread over a few hours of MLT. This demonstrates the strength of our technique, in the ability to resolve local time differences at high spatial resolution of the reconnection electric fields.

5.3. Physical Interpretation of the Reconnection Electric Field and Reconnection Voltages Estimates

For the substorm in this event, the OCB motion contributed more than convection to the estimated magnitude of E_r . As shown in Figure 7d, Φ_N (blue solid line) follows the shape of E_r estimates assuming zero convection (cyan line). Convection contributed only very little to Φ_N . Lam et al. (2006), in contrast, found that both OCB velocities

and ionospheric convection across a stationary OCB contributed to the reconnection voltage during an isolated substorm.

Figure 9a shows for reference the reconnection voltage calculated from OCB motion and ionospheric convection separately, together with the total reconnection voltage (same as Figure 7d). Figure 9b shows the relative contributions of the OCB motion and ionospheric velocity to Φ_N . During the pseudobreakup near 06:00 UT, the convection velocity was the primary contributor to Φ_N , while the OCB was nearly stationary (see Figure 5). In contrast, during the substorm, the reconnection voltage was dominated by poleward motion of the OCB.

Over longer timescales, such as over the course of a full Dungey cycle, the average reconnection electric field associated with OCB motion is nearly zero. Therefore, it is the average ionospheric convection that is primarily responsible for producing a non-zero reconnection electric field. One can get a feeling for this effect in Figure 7. The estimates of Φ_N due to the OCB (Figure 7c and cyan trace in Figure 7d) vary from negative to positive over the course of a substorm and pseudobreakup, and average to roughly zero. By contrast, estimates of Φ_N due to convection (Figure 7b) are primarily positive for the entire interval.

As shown in Figure 7b, there is no sustained increase in the ionospheric velocity despite high reconnection electric fields (Figure 7a), indicating that the ionospheric response to magnetotail reconnection is delayed. This finding agrees with Grocott et al. (2009), who showed that for substorm onsets between approximately 64° and 66° MLAT (our onset is at 65.7° MLAT) ionospheric convection does not immediately pick up after substorm onset. They suggested that high conductivities suppress the convection, via collisions/friction with neutrals, and delay the ionospheric response to increased earthward convection in the magnetotail. A delay in the ionospheric convection after substorm onset also implies that the observed reconnection electric fields are due to reconfiguration of magnetic field lines from an open to a closed configuration. A tailward propagating X-line is consistent with the rapid poleward expansion shown in Figures 5f, 6, and 7c. What the difference between pseudobreakup and substorm behaviors may correspond to in the tail is an interesting yet open question (A. Aikio et al., 1999; Nakamura et al., 1994; Rostoker, 1998). Nevertheless, our estimates support the idea that tail reconnection activity associated with a pseudobreakup is weaker than that associated with a substorm.

One last remark with respect to interpreting reconnection electric field estimates when using SuperDARN observations regards the absorption of backscatter echoes due to enhanced particle precipitation. Provan et al. (2004) observed a decrease in SuperDARN backscatter echoes for at least 30 min after substorm onset. This leads to fewer backscatter echoes in the Lompe inversion. However, this highlights the strength of using an assimilative model, as we mitigate this effect by including ground magnetometers in the inversion, and therefore believe that the observed trends in Figures 7a and 7b are not the result of this instrumental limitation. Relying more on ground magnetometers for describing ionospheric convection is associated with larger uncertainties since convection estimates are related to ground magnetometers through conductances via the ionospheric Ohm's law.

The event studied in this paper shows that the auroral oval can be very dynamic and both contraction and expansion of the polar cap can occur at the same time at different magnetic local times. This is a consequence of the fact that reconnection occurs in the magnetotail both at different times and at different locations. After substorm onset the OCB moved poleward in the dusk/pre-midnight sector and equatorward in the post-midnight/dawn sector (time steps T_9 – T_{14} in Figure 6). Approximately 40 min after substorm onset the OCB moved poleward both at dusk and dawn as shown in Figure 6 and the reconnection site propagated to other MLT sectors (Figure 8a), also shown by Blanchard et al. (1997) for a different event. This asymmetric behavior of the OCB motion might indicate that the newly closed field lines at dusk affect the location of the open field lines in the polar cap at a different MLT. The equatorward motion of the OCB is not always balanced by our estimates of the ionospheric convection, and smaller convection velocities projected onto the OCB normal lead to apparent negative reconnection electric fields.

6. Conclusions

Estimation of nightside reconnection electric fields from ionospheric measurements represents a powerful remote sensing technique for magnetotail dynamics. Calculating the uncertainty of the reconnection electric field E_r and reconnection voltage Φ_N is challenging, but is nevertheless crucial both for assessing the reliability of our estimates of E_r and Φ_N , and for understanding what is required to improve these estimates. This study is the first to estimate these uncertainties over such a large area using nearly all available information. With the Lompe data

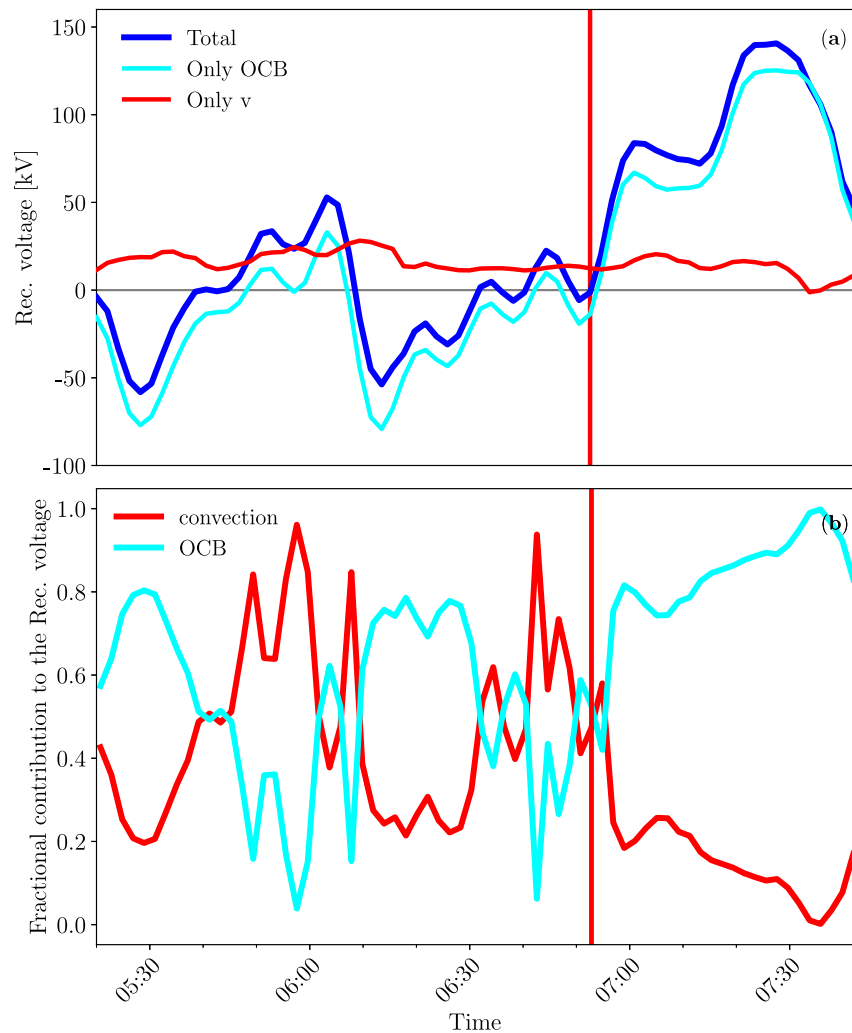


Figure 9. (a) Shows the different contributions to the total reconnection voltage, same as Figure 7d. (b) Shows the fractional contribution to the total reconnection electric field from the open-closed boundary (cyan) and from the ionospheric convection velocity (red).

assimilation technique together with the OCB estimates from global UV imaging, we have traced the evolution of the reconnection electric fields and their development in time and space at high resolution, with quantitative uncertainties. The main findings of this study are the following:

- Uncertainties in Φ_N calculated using the Lompe model variance have mean values of $\sim\pm 16$ and $\sim\pm 36$ kV for the conductance method, roughly 11% and 26%, respectively, of the reconnection voltage observed during the peak of substorm expansion phase. The uncertainty based on conductances has a much larger standard deviation (9 kV) than the variance-based uncertainty (1 kV).
- The uncertainties in the reconnection electric field are $\sim\pm 2$ and $\sim\pm 7$ mV/m at the peak of substorm expansion, for the Lompe model variance and the conductance method, respectively, roughly 4% and 13% of the peak reconnection electric field.
- We quantitatively show that the main contributor to E_r estimates after substorm onset is the motion of the OCB.
- We quantitatively show that during the pseudobreakup the main contributor to E_r estimates is the ionospheric convection, as the OCB is approximately stationary during this period.
- Unphysical negative reconnection electric fields can be used as an indication of the order of magnitude (approximately 50%, ~ 30 kV) of missing sources of uncertainty that still need to be taken into account in order to reconcile our estimates of the reconnection electric field with present understanding of magnetic reconnection.

We believe the methods we have outlined in this paper to be a valuable tool to obtain a holistic picture of reconnection electric fields and their associated uncertainties. The method for deriving uncertainties in conductances could be in the future applied to auroral imagery data from the upcoming Solar wind Magnetosphere Ionosphere Link Explorer (SMILE) mission (Raab et al., 2016) and large-scale ground-based multispectral imaging such as from the Transition Region Explorer.

Appendix A: Conductance Uncertainties: Propagation of Uncertainties

The propagation of uncertainties is done following the variance propagation formula (Ku, 1966) which assumes the variables we are treating are approximately Gaussian:

$$\sigma_{f(x,y,\dots)} = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 \sigma_x^2 + \left(\frac{\partial f}{\partial y}\right)^2 \sigma_y^2 + \dots} \quad (\text{A1})$$

Including the term related to the correlation between variables Equation A1 becomes, for addition and ratio, respectively (Taylor & Thompson, 1982):

$$f = aA + bB \rightarrow \sigma_f = \sqrt{(a\sigma_A)^2 + (b\sigma_B)^2 + 2ab\sigma_{AB}}, \quad (\text{A2})$$

$$f = \frac{A}{B} \rightarrow \sigma_f = |f| \sqrt{\left(\frac{\sigma_A}{A}\right)^2 + \left(\frac{\sigma_B}{B}\right)^2 - 2 \frac{\sigma_{AB}}{AB}}, \quad (\text{A3})$$

where $\sigma_{AB} = \rho_{AB}\sigma_A\sigma_B$ and ρ_{AB} is the correlation coefficient between the variables A and B.

A1. Uncertainties on the Electron Mean Energy

$\bar{E}$ is a function of the ratio R between corrected WIC intensity I_{WIC}^* and corrected S113 intensity I_{S113}^* . Propagation of error then gives an uncertainty for $\bar{E}$

$$\sigma_{\bar{E}} = \sqrt{\left(\frac{\partial \bar{E}(R)}{\partial R}\right)^2 \sigma_R^2 + \left|\frac{\partial \bar{E}(R)}{\partial R}\right| \sigma_R}, \quad (\text{A4})$$

where $\frac{\partial \bar{E}(R)}{\partial R}$ is the derivative of the $\bar{E}$ with respect to R (found from Tables IV and VII in Frey et al. (2003)), and σ_R is the uncertainty on the ratio. A few steps are required in order to have the uncertainty on the ratio σ_R , which is derived using Equation A3

$$\sigma_R = \sqrt{\left(\frac{1}{I_{S113}^*}\right)^2 \sigma_{I_{WIC}^*}^2 + \left[\frac{I_{WIC}^*}{(I_{S113}^*)^2}\right]^2 \sigma_{I_{S113}^*}^2 - \frac{2\rho_{I_{WIC}^* I_{S113}^*} \sigma_{I_{WIC}^*} \sigma_{I_{S113}^*}}{(I_{S113}^*)^3}}. \quad (\text{A5})$$

The covariance term must be included because I_{WIC}^* and I_{S113}^* covary. We use $\rho_{WIC^* S113^*} = 0.892$, which we obtained by calculating the correlation coefficient between I_{WIC}^* and I_{S113}^* at all times and calculating the median.

To propagate the ratio uncertainty we need the uncertainties of I_{WIC}^* and I_{S113}^* counts. In the following we address I_{WIC}^* , but the procedure is precisely analogous for I_{S113}^* . The uncertainty in I_{WIC}^* is:

$$\sigma_{I_{WIC}^*} = \frac{\sqrt{I_{WIC} + \sigma_{I_{WIC}^d}^2 + \sigma_{I_{WIC}^p}^2}}{\sqrt{3}}, \quad (\text{A6})$$

where the right-hand side of Equation A6 is divided by $\sqrt{3}$ since we average the images over 3 time steps to get the conductance estimates, $\sqrt{I_{WIC}}$ is the uncertainty in I_{WIC} , $\sigma_{I_{WIC}^d}$ is the uncertainty in I_{WIC} of the dayglow given by Ohma et al. (2023) algorithm, and

$$\sigma_{I_{WIC}^p} = \sqrt{\left(\frac{\partial f_{WIC}(E_p)}{\partial E_p} F_p\right)^2 \sigma_{E_p}^2 + (f_{WIC}(E_p))^2 \sigma_{F_p}^2} \quad (A7)$$

is the uncertainty on the proton counts in WIC. Here F_p is the proton flux as measured by SI12, σ_{F_p} its associated uncertainty, and $f_{WIC}(E_p)$ is the counts in WIC in response to 1 mW/m² proton energy flux given the proton characteristic energy. We assume $E_p = 2$ keV, and $f_{WIC}(E_p)$ is given by Table V in Frey et al. (2003). The derivative $\frac{\partial f_{WIC}(E_p)}{\partial E_p}$ is calculated numerically. We assume no uncertainty in the proton characteristic energy ($\sigma_{E_p} = 0$ keV). The uncertainty in the proton flux is given by:

$$\sigma_{F_p} = \sqrt{\left(\frac{1}{f_{SI12}^*(E_p)}\right)^2 \sigma_{I_{SI12}^*}^2 + \left(\frac{I_{SI12}^*}{f(E_p)} \frac{\partial f_{SI12}^*(E_p)}{\partial E_p}\right)^2 \sigma_{E_p}^2}, \quad (A8)$$

where $f_{SI12}^*(E_p)$ is the number of counts registered by SI12 in response to 1 mW/m² proton energy flux for the 2-keV proton characteristic energy given in Table III in Frey et al. (2003), and $\frac{\partial f_{SI12}^*(E_p)}{\partial E_p}$ is derived numerically. The uncertainty after dayglow and proton removal is $\sigma_{I_{SI12}^*} = \frac{\sqrt{I_{SI12}^2 + \sigma_{SI12}^2}}{\sqrt{3}}$, where $\sqrt{I_{SI12}}$ is the uncertainty on SI12 counts prior to dayglow and proton removal and σ_{SI12}^2 is the uncertainty in I_{SI12} of the dayglow given by Ohma et al. (2023) algorithm.

A2. Uncertainties on the Energy Fluxes

We estimate F_E using the predicted WIC counts $f_{WIC}^*(E)$ for 1 mW/m² electron energy flux for the electron mean energy values found in our study $F_E \left[\frac{mW}{m^2}\right] = \frac{I_{WIC}^*}{f_{WIC}^*(E)}$ and we propagate this ratio according to Equation A1 which leads to:

$$\sigma_{F_E} = \sqrt{\left(\frac{1}{f_{WIC}^*(E)}\right)^2 \sigma_{I_{WIC}^*}^2 + \left(\frac{I_{WIC}^*}{(f_{WIC}^*(E))^2} \left(\frac{\partial f_{WIC}^*(E)}{\partial E}\right)\right)^2 \sigma_{E}^2}, \quad (A9)$$

where $f_{WIC}^*(E)$ is the counts response for I_{WIC}^* for a given electron mean energy and 1 mW/m² electron energy flux and $\frac{\partial f_{WIC}^*(E)}{\partial E}$ its derivative also given from Table IV in Frey et al. (2003).

A3. Uncertainties on Hall and Pedersen Conductances

With the expressions given in the previous subsections we now calculate the uncertainties in the Hall and Pedersen conductances (Equations 2 and 3):

$$\sigma_{\Sigma_p} = \sqrt{\left(\frac{-40(\bar{E}^2 - 16)}{(16 + \bar{E}^2)^2}\right)^2 \sqrt{F_E}^2 \sigma_E^2 + \left(\frac{40\bar{E}}{16 + \bar{E}^2}\right)^2 \left(\frac{1}{2\sqrt{F_E}}\right)^2 \sigma_{F_E}^2}, \quad (A10)$$

$$\sigma_{\Sigma_H} = \sqrt{\left[\frac{18\bar{E}^{0.85}}{16 + \bar{E}^2} \left(1.85 - \frac{2\bar{E}^2}{16 + \bar{E}^2}\right) \sqrt{F_E}\right]^2 \sigma_E^2 + \left(\frac{9\bar{E}^{1.85}}{16 + \bar{E}^2}\right)^2 \left(\frac{1}{\sqrt{F_E}}\right)^2 \sigma_{F_E}^2}. \quad (A11)$$

Appendix B: Uncertainty of the OCB Velocity

Using Equation A1 to propagate Equations 12 and 13 we obtain

$$\sigma_{u_\theta} = \sqrt{\left(\frac{u_\theta}{r}\right)^2 \sigma_r^2 + \left(\frac{u_\theta}{\partial \theta_b / \partial t}\right)^2 \sigma_{\frac{\partial \theta_b}{\partial t}}^2 + \left[2u_\theta \cot \theta_b \left(1 - \frac{u_\theta}{r \partial \theta_b / \partial t}\right)\right]^2 \sigma_{\theta_b}^2 + \left(\frac{2u_\theta \frac{\partial \theta_b}{\partial \phi}}{\sin^2 \theta_b + \left(\frac{\partial \theta_b}{\partial \phi}\right)^2}\right)^2 \sigma_{\frac{\partial \theta_b}{\partial \phi}}^2} \quad (\text{B1})$$

where $\sigma_r = 0$.

$$\sigma_{u_\phi} = \sqrt{\left(\frac{u_\phi}{r}\right)^2 \sigma_r^2 + \left(\frac{u_\phi}{\partial \theta_b / \partial t}\right)^2 \sigma_{\frac{\partial \theta_b}{\partial t}}^2 + \left[u_\phi \cot \theta_b \left(1 + \frac{2 \frac{\partial \theta_b}{\partial \phi}}{\sin \theta_b \frac{\partial \theta_b}{\partial t}} \frac{u_\phi}{r}\right)\right]^2 \sigma_{\theta_b}^2 + \left(\frac{u_\phi}{\frac{\partial \theta_b}{\partial \phi}} + \frac{2}{r \frac{\partial \theta_b}{\partial t} \sin \theta_b} u_\phi^2\right)^2 \sigma_{\frac{\partial \theta_b}{\partial \phi}}^2} \quad (\text{B2})$$

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Data Availability Statement

A processed subset of the IMAGE FUV data, the open-closed boundary identification and the SuperDARN data used in this study is available on Zenodo (Gasparini & Reistad, 2023), as is the code we have used for estimating the OCB velocity and its uncertainty (Hatch, 2023). SuperMAG ground magnetometer data are available at <https://supermag.jhuapl.edu/>. The LasPInteractive Solar Irradiance Datacenter for the F10.7 measurements can be found at <https://lasp.colorado.edu/lisird/>. The OMNI data were obtained from the GSFC/SPDF OMNIWeb interface at <https://omniweb.gsfc.nasa.gov/>. The Lompe code and examples on how to run it and instructions on how to install it are available at <https://github.com/klaundal/lompe>.

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